

Indian Statistical Institute
BSDS IInd Year
Academic Year 2025–2026: Semester I
Course: Probability II
Instructor: Antar Bandyopadhyay

Date: November 14, 2025

Time: 10:30 AM – 10:50 AM

Quiz # 4

Total Points: 10

Duration: 20 minutes

Note:

- Please write your name and roll number in the space provided.
- There is only one problem carrying a total of 10 points.
- Do all your work ONLY in the space provided.
- This is a closed note and closed book examination.

Name: _____

Roll No: _____

Problem 1. [10 points]

Show that if $X_n \sim \chi_n^2$, then

$$\frac{X_n - n}{\sqrt{n}} \xrightarrow{d} \text{Normal}(0, c^2).$$

Find the value of c .